

EXP. NO: 05
Date : 24/02/20

Aim:
To verify
for the given

Apparatus

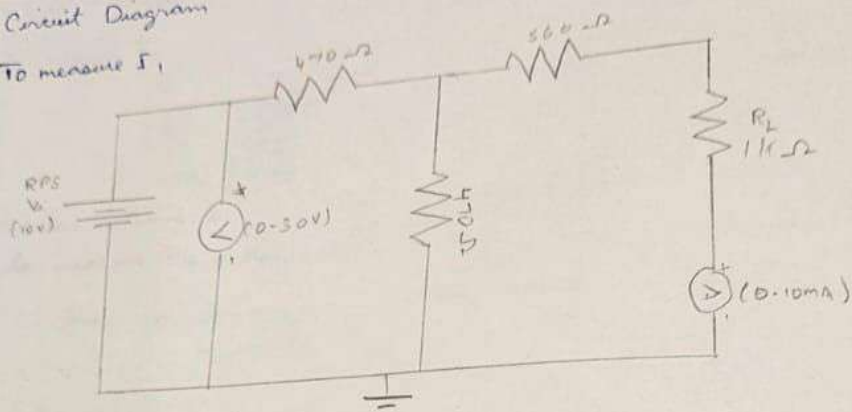
SL. NO	AP
01	Resistor
02	
03	
04	
05	BA
06	M

Procedure:

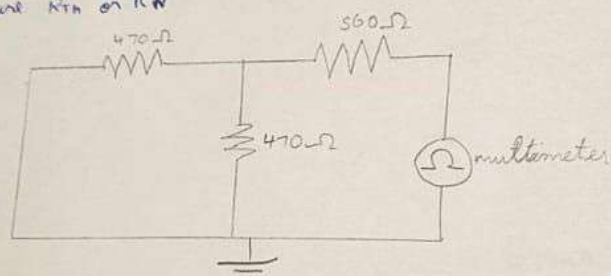
1. Make the
2. Vary the
3. Note down
4. Draw the
5. Theoretical

Circuit Diagram

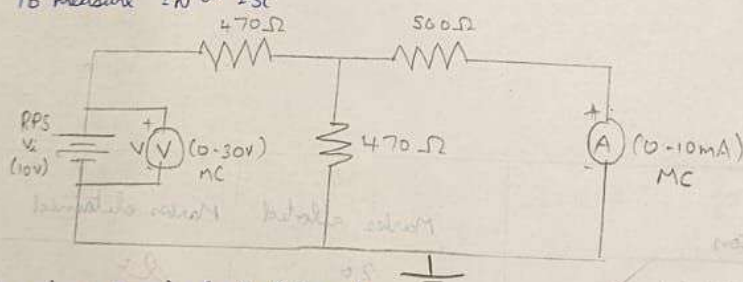
To measure I_1



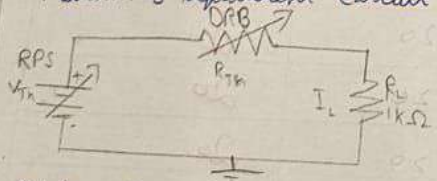
To measure R_{th} or R_N



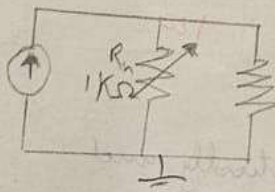
To measure I_N or I_{sc}



Thevenin's Equivalent Circuit



Norton's Equivalent Circuit



Sl. No	Description	Qty
01	Resistor 470Ω	10
02	Resistor 560Ω	10
03	Resistor 1kΩ	10
04	DC Source (RPS)	10
05	DC Source (Vth)	10
06	DC Source (IN)	10
07	DC Source (Vth)	10
08	DC Source (IN)	10
09	DC Source (Vth)	10
10	DC Source (IN)	10

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Exp. No: 05
Date: 24/02/26 Verification of Norton's theorem

Aim:

To verify Norton's theorem practically and theoretically for the given DC circuit

Apparatus required:

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
01	Regulated power supply	(0-30V)	1
02	Voltmeter	(0-30V) MC	1
03	Ammeter	(0-10mA) MC	1
04	Resistors	470 Ω , 560 Ω , 1K Ω	2, 1, 1
05	Bread board	-	1
06	Multimeter	-	1

Procedure:

1. Make the connections
2. Vary the RPS
3. Note down the readings
4. Draw the Thevenin's & equivalent circuit
5. Theoretically verify Norton's theorem

Tabular Column 1:

to measure I_L

V_i	I_L
1	0.35
2	0.75
4	1.07
6	1.45
10	1.76

Tabular Column 2:

to measure R_{TH} or R_N

V_i	$(I_N)^{R_{TH}}$
2	7.90
4	7.98
6	8.03
8	8.06
10	8.00

Tabular Column 3:

to measure I_N or I_{SC}

V_i	I_N
2	0.20
4	1.62
6	2.45
8	3.25
10	4.05

Model Calculation:

Practical value of I_L (from tabulation 1) = 2.3 mA

Verification of Norton's theorem

$$I_L = I_N \cdot R_N / (R_N + R_L) = 2.43 \text{ mA}$$

Given:

$$V_i = 10 \text{ V}$$

$$R_1 = 470 \Omega, R_2 = 470 \Omega, R_3 = 566$$

V_{TH} :

Parallel Resistance

$$R_p = 470 // 470$$

$$= 235 \Omega$$

$$V_{TH} = \frac{10 \times 235}{705} = 3.33$$

R_{TH} :

$$R_{TH} = 566 + 235$$

$$= 795 \Omega$$

I_N :

$$I_N = V_{TH} / R_{TH}$$

$$= 3.33 / 795$$

$$= 4.18 \text{ mA}$$

I_L :

$$I_L = V_{TH} / (R_{TH} + R_L)$$

$$= \frac{3.3}{1795} = 1.85 \text{ mA}$$

S.no	Description	Marks Alloted	Marks obtained
01	Experimental Setups	20	20
02	Observation / Reading	20	20
03	Calculation / Result	20	20
04	Viva	20	20
05	Record	20	20
	Total	100	100

Result :
 Thus Thevenin's and Norton's theorem is verified practically and theoretically